

P8109 Statistical Inference – Homework 4

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Problem 1

Consider sampling from the parent population with density

$$f(x | \theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

Based on a random sample X_1 of size $n = 1$ from this parent population, what is the power of the most powerful (MP) test of $H_0 : \theta = 1$ vs. $H_1 : \theta = 2$ if $P(\text{Type I error}) = \alpha = 0.05$?

Under H_0 , $f(x | 1) = 1$ (so $X_1 \sim \text{Uniform}(0, 1)$); under H_1 , $f(x | 2) = 2x$. By the Neyman–Pearson Lemma, the MP test rejects H_0 when the likelihood ratio

$$\Lambda(x_1) = \frac{f(x_1 | 2)}{f(x_1 | 1)} = 2x_1$$

is large. Since Λ is strictly increasing in x_1 , the rejection region is of the form

$$R = \{x_1 : x_1 \geq c\}$$

for some constant $c \in (0, 1)$.

Under H_0 , $X_1 \sim \text{Uniform}(0, 1)$, so

$$\alpha = P_{\theta=1}(X_1 \geq c) = 1 - c = 0.05 \implies c = 0.95.$$

Under H_1 , $f(x | 2) = 2x$, so

$$\text{Power} = P_{\theta=2}(X_1 \geq 0.95) = \int_{0.95}^1 2x \, dx = [x^2]_{0.95}^1 = 1 - 0.9025 = 0.0975.$$

Problem 2

Consider sampling from the parent population with density

$$f(x | \theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

If X_1 and X_2 constitute a random sample of size $n = 2$, show that the rejection region of size α for the MP test of $H_0 : \theta = 1$ vs. $H_1 : \theta = 2$ is $X_1 X_2 \geq k^*$, where $k^* - k^* \log k^* = 1 - \alpha$.

Under independence the joint densities are

$$L(1 | x_1, x_2) = 1, \quad L(2 | x_1, x_2) = 4x_1 x_2.$$

Hence

$$\Lambda(x_1, x_2) = \frac{L(2 | x_1, x_2)}{L(1 | x_1, x_2)} = 4x_1x_2,$$

which is a strictly increasing function of the sufficient-like statistic $T = X_1X_2$. By the Neyman–Pearson Lemma, the MP rejection region has the form

$$R = \{(x_1, x_2) : x_1x_2 \geq k^*\}.$$

Under $\theta = 1$, $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$, so $-\log X_i \stackrel{\text{iid}}{\sim} \text{Exponential}(1)$ and

$$W = -\log X_1 - \log X_2 \sim \text{Gamma}(2, 1),$$

with density $f_W(w) = we^{-w}$, $w > 0$, and (by the Erlang tail / Poisson relationship)

$$P(W \geq w) = e^{-w}(1 + w), \quad w \geq 0.$$

The event $\{X_1X_2 \geq k^*\}$ is equivalent to $\{W \leq -\log k^*\}$. Setting the size equal to α ,

$$\alpha = P_{H_0}(X_1X_2 \geq k^*) = 1 - P(W \geq -\log k^*) = 1 - e^{\log k^*}(1 - \log k^*) = 1 - (k^* - k^* \log k^*),$$

so k^* satisfies

$$k^* - k^* \log k^* = 1 - \alpha,$$

which is the stated form.

Problem 3

Let $Y_1, \dots, Y_n$ be a random sample from the density

$$f(y | \theta) = \frac{1}{2}(1 + \theta y), \quad -1 < y < 1, \quad \theta > -1.$$

- (a) Under H_0 , $f(y | 0) = 1/2$ on $(-1, 1)$ (so the Y_i are $\text{Uniform}(-1, 1)$); under H_1 , $f(y | 1) = (1 + y)/2$. By the NP Lemma the MP test rejects for large values of

$$\Lambda(\mathbf{y}) = \frac{\prod_{i=1}^n \frac{1}{2}(1 + y_i)}{\prod_{i=1}^n \frac{1}{2}} = \prod_{i=1}^n (1 + y_i).$$

Because each factor $(1 + y_i) > 0$ on the support, Λ is strictly monotone increasing in the statistic $T(\mathbf{Y}) = \prod_{i=1}^n (1 + Y_i)$ (equivalently in $\sum_{i=1}^n \log(1 + Y_i)$). Hence the MP rejection region is

$$R = \left\{ \mathbf{y} : \prod_{i=1}^n (1 + y_i) \geq c \right\},$$

with the constant c determined by $P_{\theta=0}(\prod_{i=1}^n (1 + Y_i) \geq c) = \alpha$.

- (b) With $n = 1$ the region reduces to $\{Y_1 \geq c\}$ (since $1 + y$ is monotone in y). Under H_0 , $Y_1 \sim \text{Uniform}(-1, 1)$ with density $1/2$, so

$$0.05 = P_{\theta=0}(Y_1 \geq c) = \int_c^1 \frac{1}{2} dy = \frac{1-c}{2}.$$

Solving, $1 - c = 0.10$, so

$$c = 0.90.$$

- (c) Under H_1 the density is $f(y | 1) = (1 + y)/2$, so

$$\text{Power} = P_{\theta=1}(Y_1 \geq 0.9) = \int_{0.9}^1 \frac{1+y}{2} dy = \left[\frac{y}{2} + \frac{y^2}{4} \right]_{0.9}^1 = \left(\frac{1}{2} + \frac{1}{4} \right) - (0.45 + 0.2025).$$

That is,

$$\text{Power} = 0.75 - 0.6525 = 0.0975.$$

Problem 4

Suppose $X \sim \text{Poisson}(\lambda)$. A sample of size $n = 10$ is used to test $H_0 : \lambda = 0.1$ vs. $H_1 : \lambda > 0.1$. Construct the randomized MP test of size $\alpha = 0.05$.

Let $T = \sum_{i=1}^{10} X_i$. Since $X_1, \dots, X_{10} \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$ form a one-parameter exponential family with natural statistic T , the sum T is sufficient for λ , and moreover $T \sim \text{Poisson}(n\lambda)$. Under H_0 ,

$$T \sim \text{Poisson}(n\lambda_0) = \text{Poisson}(1).$$

For any fixed $\lambda_1 > 0.1$, the joint pmf ratio

$$\frac{L(\lambda_1 | \mathbf{x})}{L(\lambda_0 | \mathbf{x})} = \exp(-n(\lambda_1 - \lambda_0)) \left(\frac{\lambda_1}{\lambda_0} \right)^T$$

is strictly increasing in T (since $\lambda_1/\lambda_0 > 1$). Hence the Poisson family has monotone likelihood ratio (MLR) in T , and by Theorem 8.1 (Lecture 8) a test that rejects for large values of T is UMP for the composite alternative $H_1 : \lambda > 0.1$ at its size. Since T is discrete, an exact size- α test requires randomization at the boundary.

We seek the integer k such that

$$P_{H_0}(T \geq k+1) \leq \alpha < P_{H_0}(T \geq k).$$

Using $T \sim \text{Poisson}(1)$:

```
alpha <- 0.05
k_vals <- 0:6
tail_ge <- ppois(k_vals - 1, lambda = 1, lower.tail = FALSE) # P(T >= k)
pmf <- dpois(k_vals, lambda = 1) # P(T = k)
data.frame(k = k_vals,
            P_T_ge_k = round(tail_ge, 6),
            P_T_eq_k = round(pmf, 6))
```

```
##   k P_T_ge_k P_T_eq_k
## 1 0 1.000000 0.367879
## 2 1 0.632121 0.367879
## 3 2 0.264241 0.183940
## 4 3 0.080301 0.061313
## 5 4 0.018988 0.015328
## 6 5 0.003660 0.003066
## 7 6 0.000594 0.000511
```

From the table,

$$P_{H_0}(T \geq 4) \approx 0.01899, \quad P_{H_0}(T \geq 3) \approx 0.08030,$$

so $P_{H_0}(T \geq 4) < 0.05 < P_{H_0}(T \geq 3)$. Therefore we take $k = 3$ and randomize on the boundary $\{T = 3\}$.

We require

$$P_{H_0}(T > 3) + \gamma P_{H_0}(T = 3) = \alpha,$$

i.e.,

$$0.018988 + \gamma \cdot 0.061313 = 0.05 \implies \gamma = \frac{0.05 - 0.018988}{0.061313} \approx 0.5058 \in [0, 1].$$

The randomized MP (UMP) test is, writing $T = \sum_{i=1}^{10} X_i$,

$$\phi(\mathbf{x}) = \begin{cases} 1, & T > 3, \\ \gamma \approx 0.5058, & T = 3, \\ 0, & T < 3. \end{cases}$$

Operationally: compute T ; if $T \geq 4$ reject H_0 ; if $T \leq 2$ do not reject; if $T = 3$, draw $U \sim \text{Uniform}(0, 1)$ independently of the data and reject iff $U < \gamma$.

By construction $E_{H_0}[\phi(\mathbf{X})] = P_{H_0}(T \geq 4) + \gamma P_{H_0}(T = 3) = \alpha$, so the test has size exactly 0.05. By the Neyman–Pearson Lemma and the MLR property of the Poisson family, this randomized test is the uniformly most powerful (and hence MP) test of size 0.05 for $H_0 : \lambda = 0.1$ against $H_1 : \lambda > 0.1$.